

## Introduction

The proposed change involves creating separate Snowflake warehouses for different departments, projects, and workloads. This strategy aims to enhance cost tracking, and ensure that production and development environments are isolated from one another. By assigning dedicated warehouses to specific teams and tasks, we can **monitor and optimize compute resource usage** more effectively.

## Goals

1. Improved Cost Tracking: Separate warehouses allow for detailed tracking of compute costs by department, project, and task, which helps in budget allocation and identifying optimization opportunities. a. may help with project estimation cost in the long run and help with the budgeting process as project proposals are assessed.
2. Flexibility and Scalability: Enable the ability to scale resources independently for different departments and workloads, providing flexibility to adjust compute power based on real-time needs.

## Advantages

- Better Cost Control: Granular tracking of compute usage allows for more accurate cost allocation, which can drive cost-saving initiatives and better budget management.
- Warehouses don't have to be permanent and can be named based on department, task, or project

## Disadvantages

- Permissions Management: Managing permissions becomes more complex as each warehouse will need access to different tables. Ensuring that the right access controls are in place for each warehouse, while restricting access to sensitive data, adds administrative overhead and increases the risk of misconfigurations.

## Proposed Warehouses

### In-house Development Warehouse

- Purpose: For internal testing, development, and ad hoc queries by the in-house team.
- Environment: Development.

### Tuva's Development Warehouse

- Purpose: For Tuva's development tasks, including testing and exploratory work.
- Environment: Development.

### Tuva's Production Warehouse

- Purpose: Dedicated to Tuva's production operations, handling live data and critical processes.
- Environment: Production.

### **Athena Project Warehouse**

- Purpose: Athena project, tracking this specific project separately.
- Environment: Project-specific (either Development or Production).

### **Production Pipelines Warehouse**

- Purpose: Handles production workloads, including Snowpark, Lambda functions, and other pipelines.
- Environment: Production.

### **Special Projects Warehouse**

- Purpose: For experiments, one-time data processing jobs, or any other special project that requires isolation and/or tracking.
- Environment: Adjustable (Development or Production).

### **Conclusion**

This structured warehouse allocation strategy aims to improve resource utilization, operational performance, and cost management while ensuring that potential drawbacks, such as permissions management and governance complexity, are addressed with proper planning and regular monitoring.